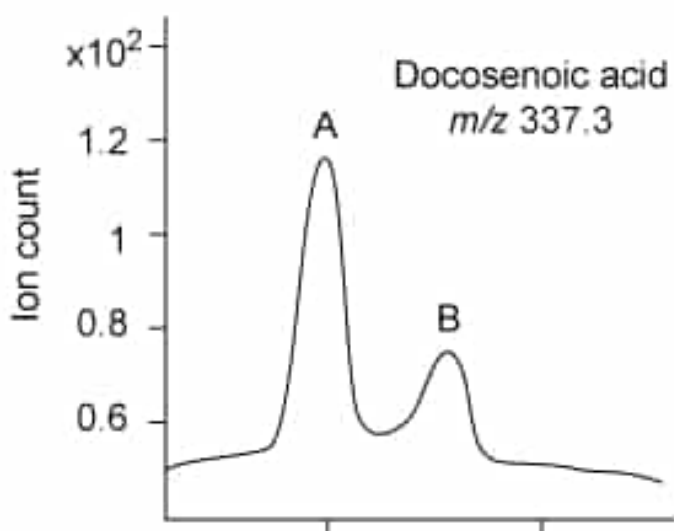


Fatty acids (FAs) are lipids that play important roles in cellular function, including energy storage and membrane structure. Most polyunsaturated fatty acids (PUFAs) are made in the body; however, omega-3 alpha-linolenic acid and omega-6 linoleic acid must be obtained from the diet. These two PUFAs are precursors to other omega-3 and omega-6 long-chain FAs that are necessary for physiological processes such as neurological development, normal growth, and immune response. Insufficient omega-3 and omega-6 long-chain FAs can lead to an increase in omega-9 fatty acids such as mead acid, and to essential fatty acid deficiency (EFAD), which can cause infertility, decreased growth, and increased susceptibility to infection.

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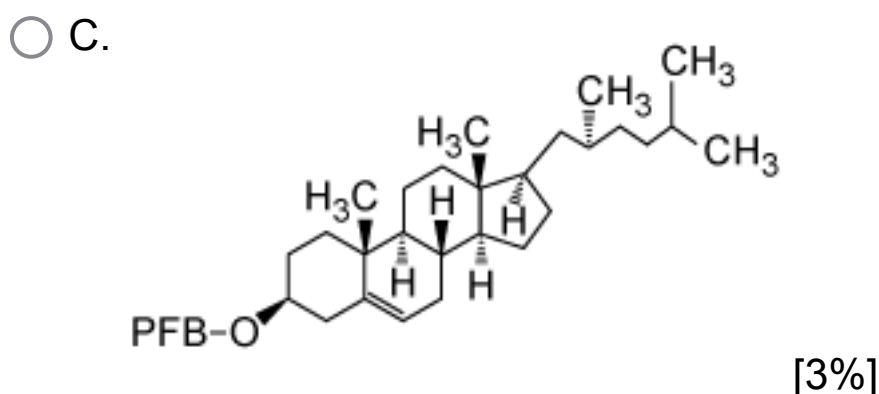
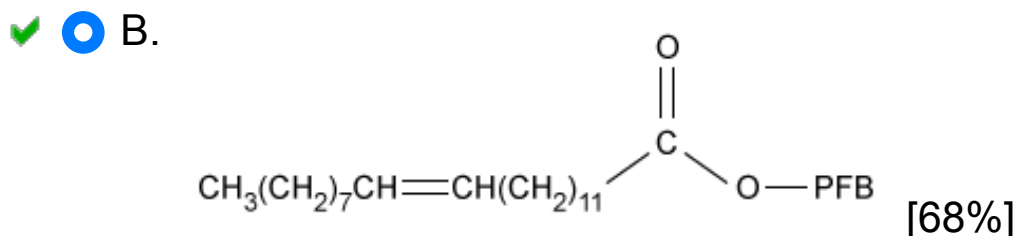
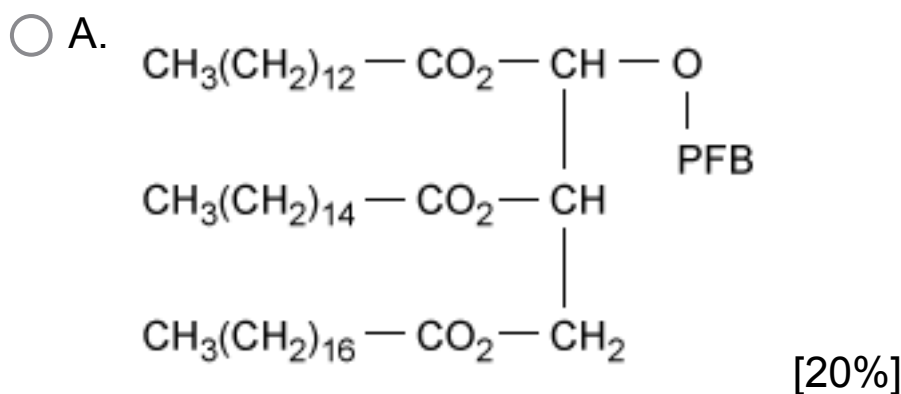


19.5 19.7
Acquisition time (min)

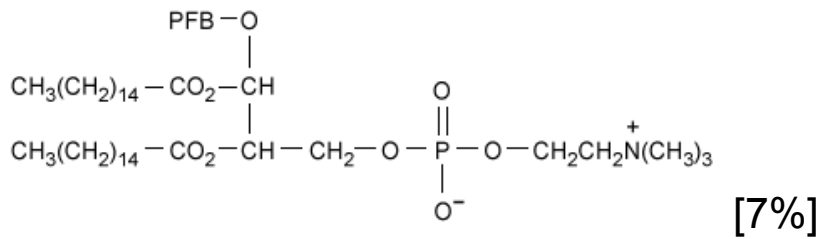
Figure 1 Selected ion chromatogram of docosenoic acid

Researchers have proposed that an elevated ratio of the triene omega-9 mead acid to the tetraene omega-6 arachidonic acid, known as the triene/tetraene (TT) ratio, can signify EFAD. Using GC and NCI-MS, they measured the TT ratio, concentration of mead acid, and individual concentrations of omega-3 and omega-6 FAs in plasma and serum samples from three patients and compared their results to normal ranges. An increased TT ratio and mead acid concentration and decreased omega-3 and omega-6 FA concentrations were observed in the patient samples compared to normal ranges.

Based on the passage, which compound was most likely analyzed by GC and NCI-MS in the experiment?



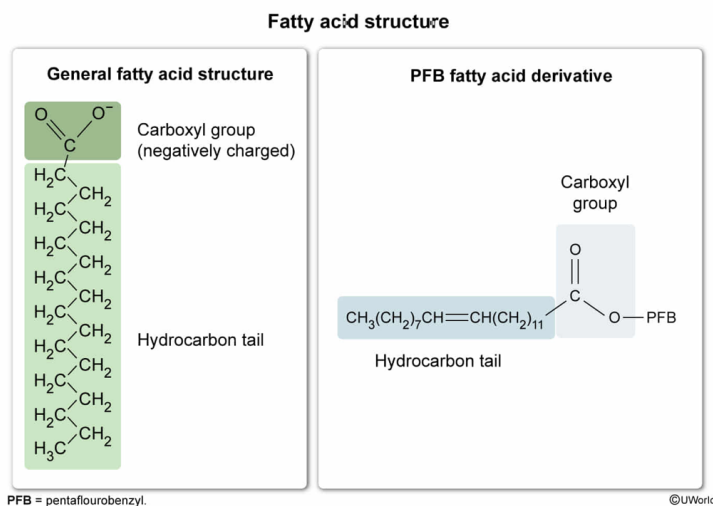
○ D.



Correct

68% answered correctly

Explanation:



Fatty acids (FAs) consist of a nonpolar **hydrocarbon chain** with a polar **carboxyl head group**, and can be classified as either **saturated or unsaturated**. Saturated FAs do not contain any double bonds in the hydrocarbon chain whereas the hydrocarbon chain of unsaturated FAs contains at least one double bond, usually in the **cis conformation**. Free FAs are usually not found in blood plasma but rather exist as a **derivative**, such as a **triglyceride** or **phospholipid**. Hydrolysis of triacylglycerides and phospholipids releases free FAs.

In the experiment, plasma and serum samples were hydrolyzed to give free FAs, which were then converted to a pentafluorobenzyl (PFB) derivative. The PFB FA derivatives were analyzed by GC and NCI-MS to determine total FA levels. Therefore, the compound with a hydrocarbon tail and a carboxyl group attached to a PFB substituent was most likely analyzed in the experiment. **Choice B** is the only compound that has these characteristics.

(Choices A and D) These compounds were derived from a triacylglyceride and the phospholipid phosphatidylcholine, respectively, both of which are present in plasma and serum samples. However, triacylglycerides and phospholipids were not analyzed directly; they were hydrolyzed to give free FAs, which were then converted to PFB derivatives and analyzed by GC and NCI-MS.

(Choice C) This compound is derived from cholesterol. Although **cholesterol** is a lipid, it contains a **steroid backbone** and is made up of isoprene units rather than FAs.

Educational objective:

Fatty acids (FAs) are made up of a nonpolar hydrocarbon chain with a polar carboxyl head group, and can be classified as either saturated or unsaturated. Free FAs are usually not found in blood plasma but rather exist as derivatives, such as a triacylglyceride or phospholipid.

Hydrolysis of triacylglycerides and phospholipids releases free FAs.

Time spent:0 sec

Subject:

Organic Chemistry

QID:402076

System:

Functional Groups and Their Reactions

Fatty acids (FAs) are lipids that play important roles in cellular function, including energy storage and membrane structure. Most polyunsaturated fatty acids (PUFAs) are made in the body; however, omega-3 alpha-linolenic acid and omega-6 linoleic acid must be obtained from the diet. These two PUFAs are precursors to other omega-3 and omega-6 long-chain FAs that are necessary for physiological processes such as neurological development, normal growth, and immune response. Insufficient omega-3 and omega-6 long-chain FAs can lead to an increase in omega-9 fatty acids such as mead acid, and to essential fatty acid deficiency (EFAD), which can cause infertility, decreased growth, and increased susceptibility to infection.

Total FA levels in plasma and serum can be used to diagnose EFAD. Plasma and serum samples from several individuals were hydrolyzed to release FAs from triacylglycerols and phospholipids. Once isolated, the free FAs were converted to pentafluorobenzyl (PFB) derivatives and separated by gas chromatography (GC). Total levels of each type of fatty acid were quantified by negative chemical ionization–mass spectrometry (NCI-MS) immediately after GC. NCI is an ionization technique that creates negatively charged ions with little to no fragmentation. The NCI-MS revealed that two of the peaks from GC had the same m/z ratio (Figure 1). They were both confirmed to be forms of an unsaturated, unbranched 22-carbon FA called omega-9 docosenoic acid.

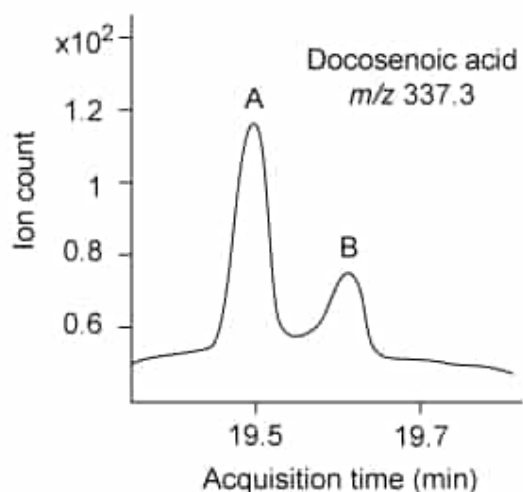
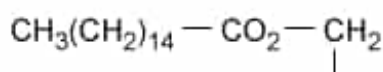
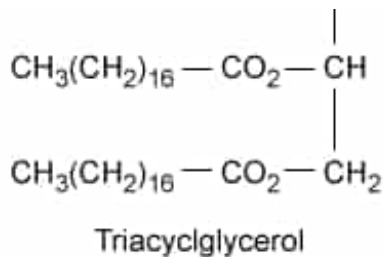


Figure 1 Selected ion chromatogram of docosenoic acid

Researchers have proposed that an elevated ratio of the triene omega-9 mead acid to the tetraene omega-6 arachidonic acid, known as the triene/tetraene (TT) ratio, can signify EFAD. Using GC and NCI-MS, they measured the TT ratio, concentration of mead acid, and individual concentrations of omega-3 and omega-6 FAs in plasma and serum samples from three patients and compared their results to normal ranges. An increased TT ratio and mead acid concentration and decreased omega-3 and omega-6 FA concentrations were observed in the patient samples compared to normal ranges.

A triacylglyceride isolated from a plasma sample is shown below.





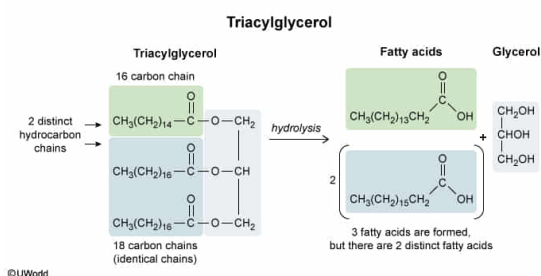
When the triacylglyceride is hydrolyzed, how many unique products are obtained other than glycerol?

- ☐ A. 0 [1%]
☐ B. 1 [9%]
☒ C. 2 [68%]
☐ D. 3 [20%]

Correct

68% answered correctly

Explanation:



Triacylglycerols are lipids that are primarily found in adipocytes for energy storage. These lipids are made up of three fatty acids attached to a glycerol molecule through **ester** linkages. **Hydrolysis** of triacylglycerols breaks the ester bonds, releasing free fatty acids and glycerol. There are three fatty acids released during the hydrolysis; however, the number of **distinct fatty acids** released **corresponds** to the number of **distinct hydrocarbon chains** in the triacylglyceride.

The triacylglycerol has two distinct hydrocarbon chains: one with 16 carbons and two with 18 carbons. The two 18-carbon chains are identical to each other and will collectively form only one product. Because there are only two distinct hydrocarbon chains, hydrolysis of this triacylglycerol will yield two distinct fatty acids in addition to glycerol.

(Choice A) Glycerol is not the only product from triacylglycerol hydrolysis. Therefore, there are additional products (ie, more than zero) from this reaction.

(Choice B) There would be one fatty acid product from triacylglycerol hydrolysis if all of the hydrocarbon chains on the triacylglycerol were identical.

(Choice D) There would be three different fatty acid products from triacylglyceride hydrolysis if all three hydrocarbon chains on the triacylglycerol were different.

Educational objective:

Triacylglycerols are made up of three fatty acids and a glycerol molecule connected through ester linkages. Hydrolysis of a triacylglycerol releases free fatty acids and glycerol. The number of distinct fatty acids released corresponds to the number of distinct hydrocarbon chains in the triacylglyceride.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402077

System:
Functional Groups and Their Reactions

Fatty acids (FAs) are lipids that play important roles in cellular function, including energy storage and membrane structure. Most polyunsaturated fatty acids (PUFAs) are made in the body; however, omega-3 alpha-linolenic acid and omega-6 linoleic acid must be obtained from the diet. These two PUFAs are precursors to other omega-3 and omega-6 long-chain FAs that are necessary for physiological processes such as neurological development, normal growth, and immune response. Insufficient omega-3 and omega-6 long-chain FAs can lead to an increase in omega-9 fatty acids such as mead acid, and to essential fatty acid deficiency (EFAD), which can cause infertility, decreased growth, and increased susceptibility to infection.

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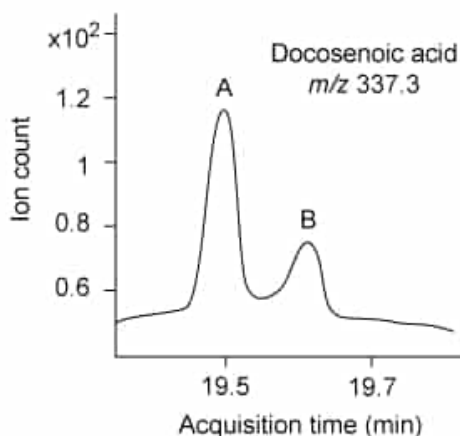


Figure 1 Selected ion chromatogram of docosenoic acid

Researchers have proposed that an elevated ratio of the triene omega-9 mead acid to the tetraene omega-6 arachidonic acid, known as the triene/tetraene (TT) ratio, can signify EFAD. Using GC and NCI-MS, they measured the TT ratio, concentration of mead acid, and individual concentrations of omega-3 and omega-6 FAs in plasma and serum samples from three patients and compared their results to normal ranges. An increased TT ratio and mead acid concentration and decreased omega-3 and omega-6 FA concentrations were observed in the patient samples compared to normal ranges.

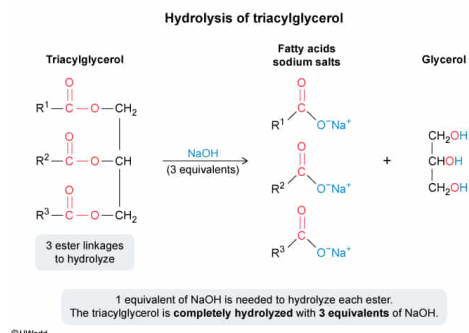
If three equivalents of NaOH are used for triacylglycerol saponification, the triacylglycerol will be:

- ✓ ☒ A. completely hydrolyzed, because one equivalent of NaOH is needed to hydrolyze each ester. [82%]
- ☐ B. completely hydrolyzed, because a catalytic amount of NaOH is needed to hydrolyze the entire triacylglycerol. [7%]
- ☐ C. partially hydrolyzed, because two equivalents of NaOH are needed to hydrolyze each ester. [5%]
- ☐ D. partially hydrolyzed, because six equivalents of NaOH are needed to hydrolyze the entire triacylglycerol. [3%]

Correct

82% answered correctly

Explanation:



Saponification is the hydrolysis of ester bonds with a strong base. **Triacylglycerols** contain three fatty acids bonded to a molecule of glycerol through ester linkages. Saponification of a triacylglycerol with a strong base, such as NaOH, releases free fatty acids (as sodium salts) and a molecule of glycerol.

One equivalent of base is defined as the amount of base needed to **hydrolyze one ester linkage** in a triacylglycerol. If three equivalents of NaOH are used to hydrolyze a triacylglycerol, the molecule will be completely hydrolyzed because each equivalent hydrolyzes one ester, and there are three esters in a triacylglycerol molecule.

(Choice B) A catalytic amount (less than one equivalent) of NaOH is not enough to hydrolyze one ester in a triacylglyceride. Therefore, a catalytic amount cannot completely hydrolyze the triacylglyceride (three esters).

(Choices C and D) Only one equivalent of base is needed to hydrolyze each ester in a triacylglyceride; therefore, two equivalents of NaOH per ester (six equivalents of NaOH total) are more than enough to completely hydrolyze the triacylglyceride.

Educational objective:

Saponification is the hydrolysis of an ester with a strong base. Triacylglycerols contain three fatty acids bonded to a molecule of glycerol through ester linkages, and saponification of a triacylglycerol with a strong base releases free fatty acids (as sodium salts) and a molecule of glycerol. One equivalent of base is needed to hydrolyze one ester linkage; therefore, three equivalents of base are needed to completely hydrolyze a triglyceride.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402078

System:
Functional Groups and Their Reactions

Fatty acids (FAs) are lipids that play important roles in cellular function, including energy storage and membrane structure. Most polyunsaturated fatty acids (PUFAs) are made in the body; however, omega-3 alpha-linolenic acid and omega-6 linoleic acid must be obtained from the diet. These two PUFAs are precursors to other omega-3 and omega-6 long-chain FAs that are necessary for physiological processes such as neurological development, normal growth, and immune response. Insufficient omega-3 and omega-6 long-chain FAs can lead to an increase in omega-9 fatty acids such as mead acid, and to essential fatty acid deficiency (EFAD), which can cause infertility, decreased growth, and increased susceptibility to infection.

Total FA levels in plasma and serum can be used to diagnose EFAD. Plasma and serum samples from several individuals were hydrolyzed to release FAs from triacylglycerols and phospholipids. Once isolated, the free FAs were converted to pentafluorobenzyl (PFB) derivatives and separated by gas chromatography (GC). Total levels of each type of fatty acid were quantified by negative chemical ionization–mass spectrometry (NCI-MS) immediately after GC. NCI is an ionization technique that creates negatively charged ions with little to no fragmentation. The NCI-MS revealed that two of the peaks from GC had the same m/z ratio (Figure 1). They were both confirmed to be forms of an unsaturated, unbranched 22-carbon FA called omega-9 docosenoic acid.

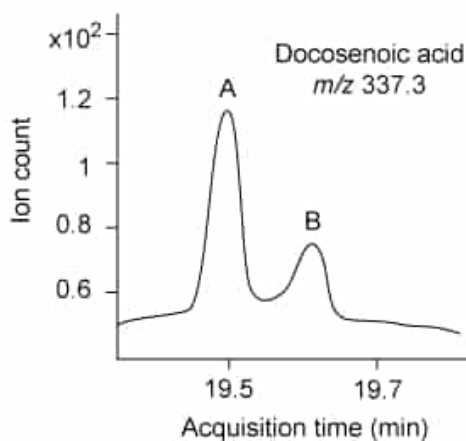


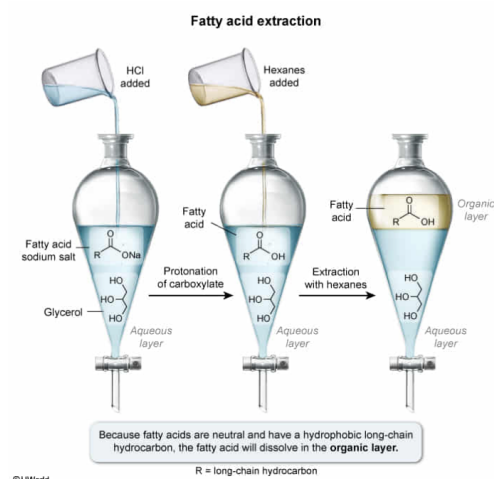
Figure 1 Selected ion chromatogram of docosenoic acid

Researchers have proposed that an elevated ratio of the triene omega-9 mead acid to the tetraene omega-6 arachidonic acid, known as the triene/tetraene (TT) ratio, can signify EFAD. Using GC and NCI-MS, they measured the TT ratio, concentration of mead acid, and individual concentrations of omega-3 and omega-6 FAs in plasma and serum samples from three patients and compared their results to normal ranges. An increased TT ratio and mead acid concentration and decreased omega-3 and omega-6 FA concentrations were observed in the patient samples compared to normal ranges.

Researchers separated fatty acid sodium salts from glycerol by adding HCl and then extracting with hexanes. Which statement correctly describes the extraction process?

- ☐ A. HCl protonates the carboxylate, and the fatty acid is found in the aqueous layer. [19%]
- ☐ B. The carboxyl group is deprotonated, and the fatty acid is dissolved in the polar solvent. [3%]
- ☒ C. HCl protonates the carboxylate, and the fatty acid is found in the organic layer. [69%]
- ☐ D. The carboxyl group is deprotonated, and the fatty acid is dissolved in the nonpolar solvent. [8%]

Explanation:



Fatty acid sodium salts are soluble in aqueous solutions because the charged carboxyl group and sodium ion interact with water through **ion-dipole interactions**. **Protonation of fatty acid** sodium salts renders the molecule insoluble in water because the fatty acid is no longer a charged species, and therefore cannot interact as readily with water. In addition, the **long-chain hydrocarbons** on fatty acids greatly **increase** the **hydrophobicity** of these molecules, causing fatty acids to be insoluble in water and **soluble** in an **organic solvent**.

Separation of glycerol from fatty acid sodium salts can be accomplished by protonation of the negatively charged carboxyl group, which can be achieved by the addition of a strong acid, such as HCl. Hexanes are nonpolar organic molecules that can readily interact with the protonated fatty acids. Therefore, protonated fatty acids will enter the organic layer after extraction with the organic solvent hexanes.

(Choice A) Although HCl does protonate the carboxylate, the fatty acid is found in the organic layer (hexanes) rather than the aqueous layer.

(Choices B and D) The carboxyl groups of fatty acid sodium salts are negatively charged and do not have protons that can be removed. Instead, the carboxyl group is *protonated* by HCl and after extraction with hexanes, the fatty acid is found in the *nonpolar* solvent (hexanes) instead of the polar solvent (water).

Educational objective:

Fatty acids sodium salts are soluble in water due to ion-dipole interactions of the charged carboxyl group and sodium ion with water. Protonation of fatty acid sodium salts renders the molecule insoluble in water because the fatty acid is no longer a charged species, and therefore cannot interact as readily with water. The long-chain hydrocarbon makes fatty acids hydrophobic, allowing for solubility in organic solvents.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402079

System:
Functional Groups and Their Reactions

Fatty acids (FAs) are lipids that play important roles in cellular function, including energy storage and membrane structure. Most polyunsaturated fatty acids (PUFAs) are made in the body; however, omega-3 alpha-linolenic acid and omega-6 linoleic acid must be obtained from the diet. These two PUFAs are precursors to other omega-3 and omega-6 long-chain FAs that are necessary for physiological processes such as neurological development, normal growth, and immune response. Insufficient omega-3 and omega-6 long-chain FAs can lead to an increase in omega-9 fatty acids such as mead acid, and to essential fatty acid deficiency (EFAD), which can cause infertility, decreased growth, and increased susceptibility to infection.

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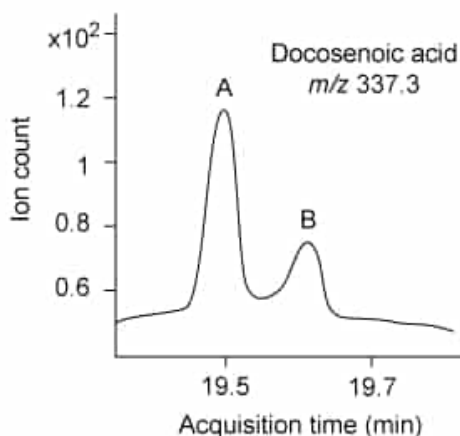


Figure 1 Selected ion chromatogram of docosenoic acid

Researchers have proposed that an elevated ratio of the triene omega-9 mead acid to the tetraene omega-6 arachidonic acid, known as the triene/tetraene (TT) ratio, can signify EFAD. Using GC and NCI-MS, they measured the TT ratio, concentration of mead acid, and individual concentrations of omega-3 and omega-6 FAs in plasma and serum samples from three patients and compared their results to normal ranges. An increased TT ratio and mead acid concentration and decreased omega-3 and omega-6 FA concentrations were observed in the patient samples compared to normal ranges.

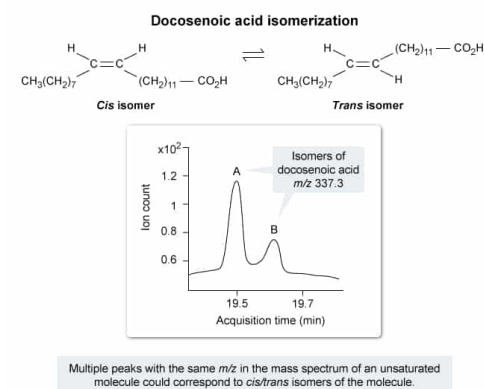
What can be inferred from Figure 1 about the relationship of Peaks A and B?

- ☐ A. Nothing can be inferred about the relationship of the peaks. [5%]
- ☐ B. The peaks are fragments of docosenoic acid. [21%]
- ☐ C. The peaks are constitutional isomers of docosenoic acid. [30%]
- ✓ ☒ D. The peaks are *cis-trans* isomers of docosenoic acid. [42%]

Correct

43% answered correctly

Explanation:



Isomers are molecules with the **same molecular formula** but **different structural arrangements**. They can be generally classified as constitutional isomers and stereoisomers. Constitutional isomers differ in atom connectivity whereas **stereoisomers** have the same atom connectivity but differ in spatial arrangement.

Stereoisomers that arise from disubstituted double bonds are called **geometric isomers**. They are classified as *cis* or *trans* based on the relative positions of the nonhydrogen substituents. Geometric isomers can be **separated by gas chromatography** because they have slightly **different boiling points**.

Docosenoic acid is an unsaturated fatty acid, meaning it contains a disubstituted double bond, and can be in the *cis* or *trans* form. Both isomers have the same molecular weight and m/z ratio, but will come off the GC column at different times, yielding two distinct peaks. Therefore, it can be inferred that Peaks A and B are *cis/trans* isomers of docosenoic acid.

(Choice A) Information is given in the passage to infer the relationship of Peaks A and B, including the fact that both peaks correspond to docosenoic acid and that docosenoic acid is unsaturated, meaning it has *cis* and *trans* forms.

(Choice B) The passage states that little to no fragmentation occurs in NCI-MS, so it is unlikely that docosenoic acid fragmented.

(Choice C) Although constitutional isomers have the same molecular weight, they have different atom connectivity, making them different compounds. The passage states that both compounds are forms of docosenoic acid, so they cannot be constitutional isomers.

Educational objective:

Isomers are molecules with the same molecular formula but different structural arrangements, and are either constitutional or stereoisomers. Disubstituted double bonds give rise to *cis* or *trans* isomers, depending on the relative positions of nonhydrogen substituents. These isomers have slightly different boiling points and can be separated by gas chromatography.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402080

System:
Functional Groups and Their Reactions

Fatty acids (FAs) are lipids that play important roles in cellular function, including energy storage and membrane structure. Most polyunsaturated fatty acids (PUFAs) are made in the body; however, omega-3 alpha-linolenic acid and omega-6 linoleic acid must be obtained from the diet. These two PUFAs are precursors to other omega-3 and omega-6 long-chain FAs that are necessary for physiological processes such as neurological development, normal growth, and immune response. Insufficient omega-3 and omega-6 long-chain FAs can lead to an increase in omega-9 fatty acids such as mead acid, and to essential fatty acid deficiency (EFAD), which can cause infertility, decreased growth, and increased susceptibility to infection.

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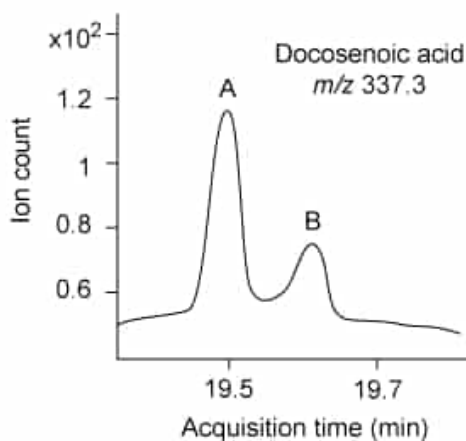


Figure 1 Selected ion chromatogram of docosenoic acid

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Which of the following conclusions is NOT consistent with the results of the patient plasma and serum sample analysis described in the passage?

- ☐ A. The ratio of mead acid to arachidonic acid observed in the patient plasma and serum samples indicates EFAD. [9%]
- ☒ B. The concentration of arachidonic acid in the patient plasma and serum samples is higher than the normal range. [61%]
- ☐ C. The concentrations of alpha-linolenic and linoleic acids in the patient plasma and serum samples are lower than the normal range. [15%]
- ☐ D. The observed mead acid concentration is a result of the decreased omega-3 and omega-6 fatty acid concentrations relative to the normal range. [13%]

Correct

61% answered correctly

Explanation:

Causes of EFAD: omega-3 & omega-6 FA deficiencies

Experimental results	Conclusion	In agreement with experimental results?
Decreased [omega-3 FA] & [omega-6 FA]	[Alpha linolenic acid] & [linoleic acid] are lower than the normal range	Yes (these are omega-3 & omega-6 FAs)
	[Arachidonic acid] in the patient plasma & serum samples is higher than the normal range	No Arachidonic acid is an omega-6 FA, so its concentration is lower than the normal range Arachidonic is in the denominator of the TT ratio. [Arachidonic acid] must decrease for TT ratio to increase
Increased [mead acid]	Observed [mead acid] is a result of the decreased [omega-3-FA] & [omega-6 FA]	Yes
Increased TT ratio	Observed TT ratio indicates EFAD	Yes

FA = fatty acid; [] = concentration; TT = triene/tetraene; EFAD = essential fatty acid deficiency.
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The passage states essential fatty acid deficiency (EFAD) is caused by a deficiency in omega-3 and omega-6 long-chain fatty acids (FAs). Omega-3 alpha-linolenic and omega-6 linoleic acids are precursors to long-chain FAs, and must be obtained from the diet. Given that the omega-3 and omega-6 FA concentrations in the patient plasma and serum samples were lower than normal ranges, alpha-linolenic and linoleic acids concentrations should be lower than the normal range (**Choice C**).

Omega-3 and omega-6 FA deficiencies lead to an increase in omega-9 FAs including mead acid, which was the triene used to measure the TT ratio. Because a decrease in omega-3 and omega-6 FA concentrations was observed in the patient plasma and serum samples, a higher than normal mead acid concentration that was higher than normal ranges was also observed in the patient plasma and serum samples (**Choice D**).

The passage also states that an elevated mead acid/arachidonic acid ratio, or TT ratio, is indicative of EFAD. An increased TT ratio was observed in the plasma and serum samples, indicating that the patients likely have EFAD (**Choice A**).

The omega-6 FA concentrations observed in the patient plasma and serum samples were lower than normal ranges, and an increased TT ratio was observed. Arachidonic acid is an omega-6 FA, and an increased concentration of arachidonic acid would decrease the TT ratio rather than increase it because arachidonic acid is in the denominator of the ratio. Therefore, the concentration of arachidonic acid is *not* expected to be higher than normal.

Educational objective:

Drawing conclusions based on experimental results requires analysis of given information. The effect that results have on another variable can be applied to make conclusions about what changes should occur to that variable. When given information about how a ratio changes, assess what increases or decreases to cause the ratio to change in the specified way.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402081

System:
Functional Groups and Their Reactions